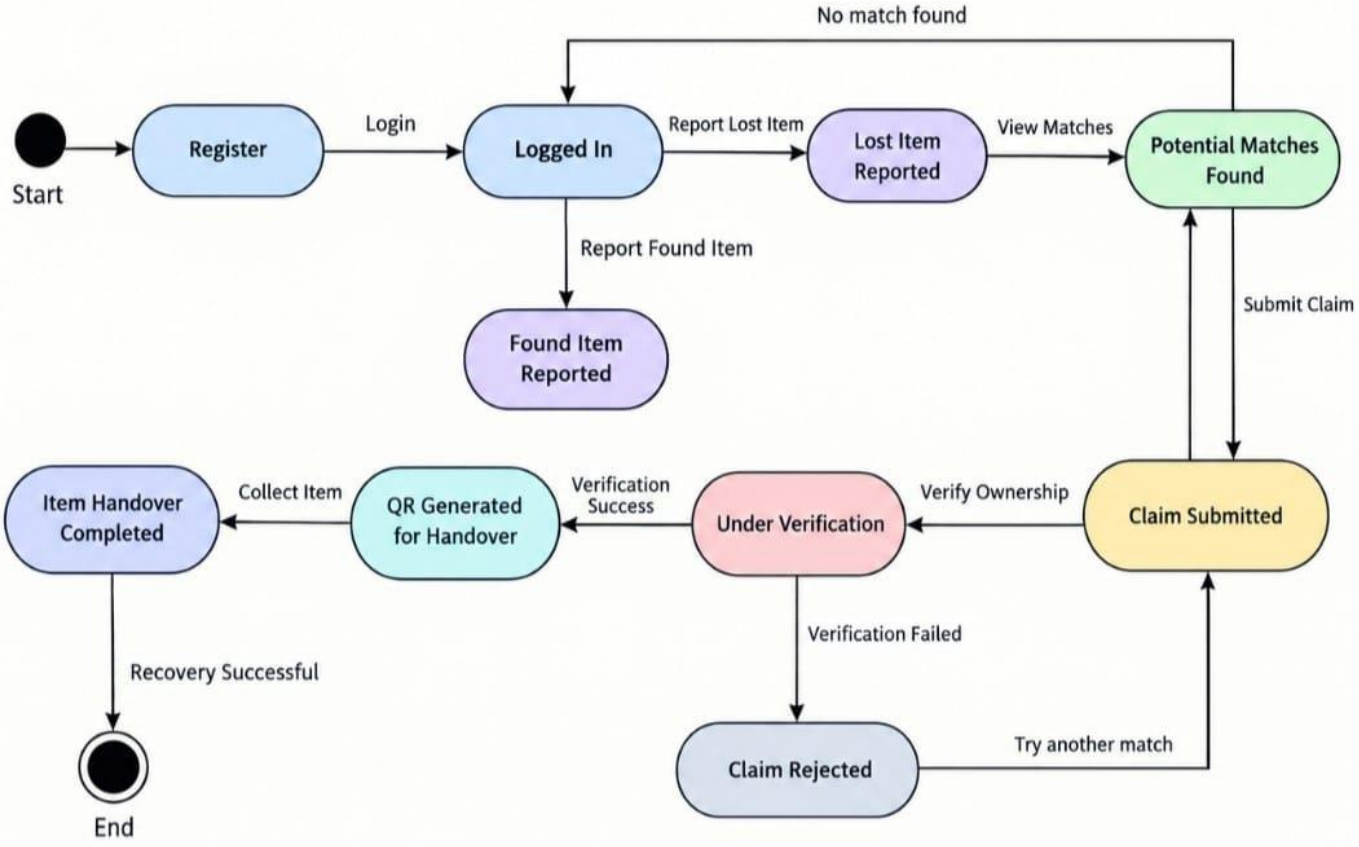
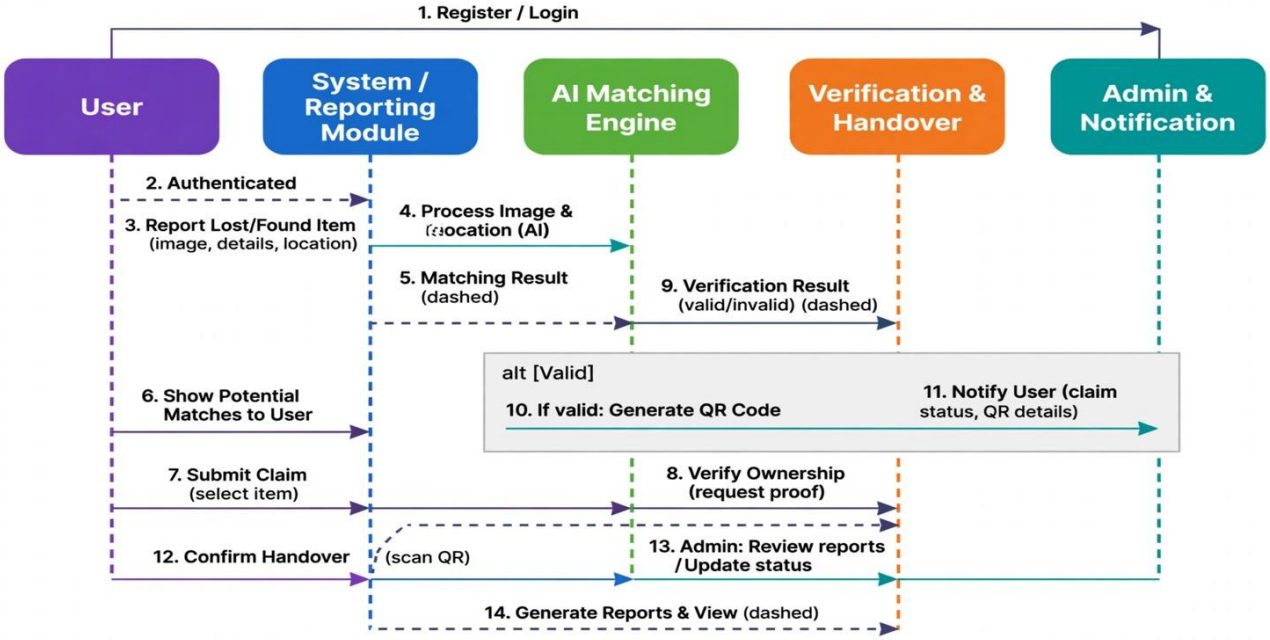
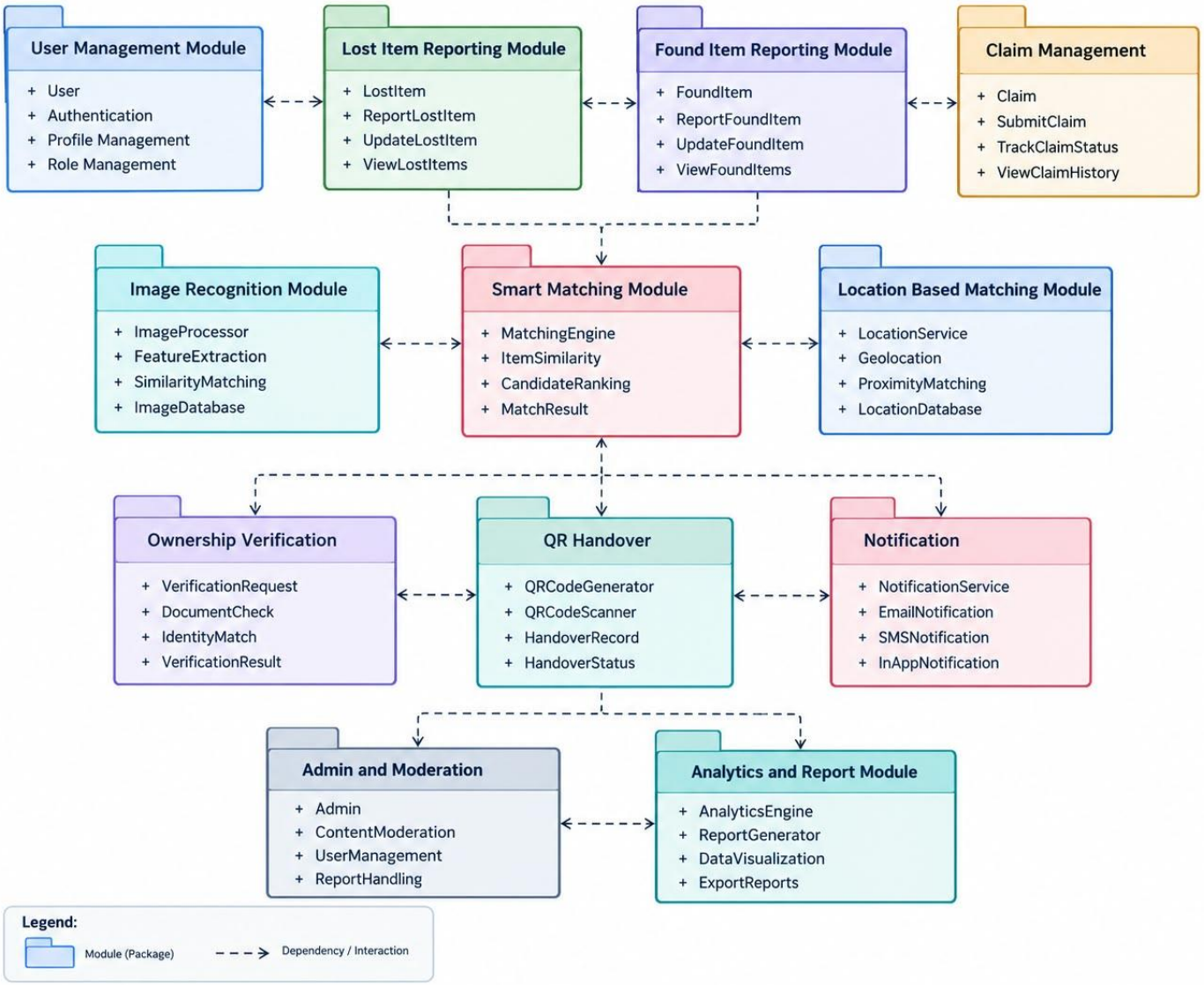
State Diagram – Lost & Found Management System

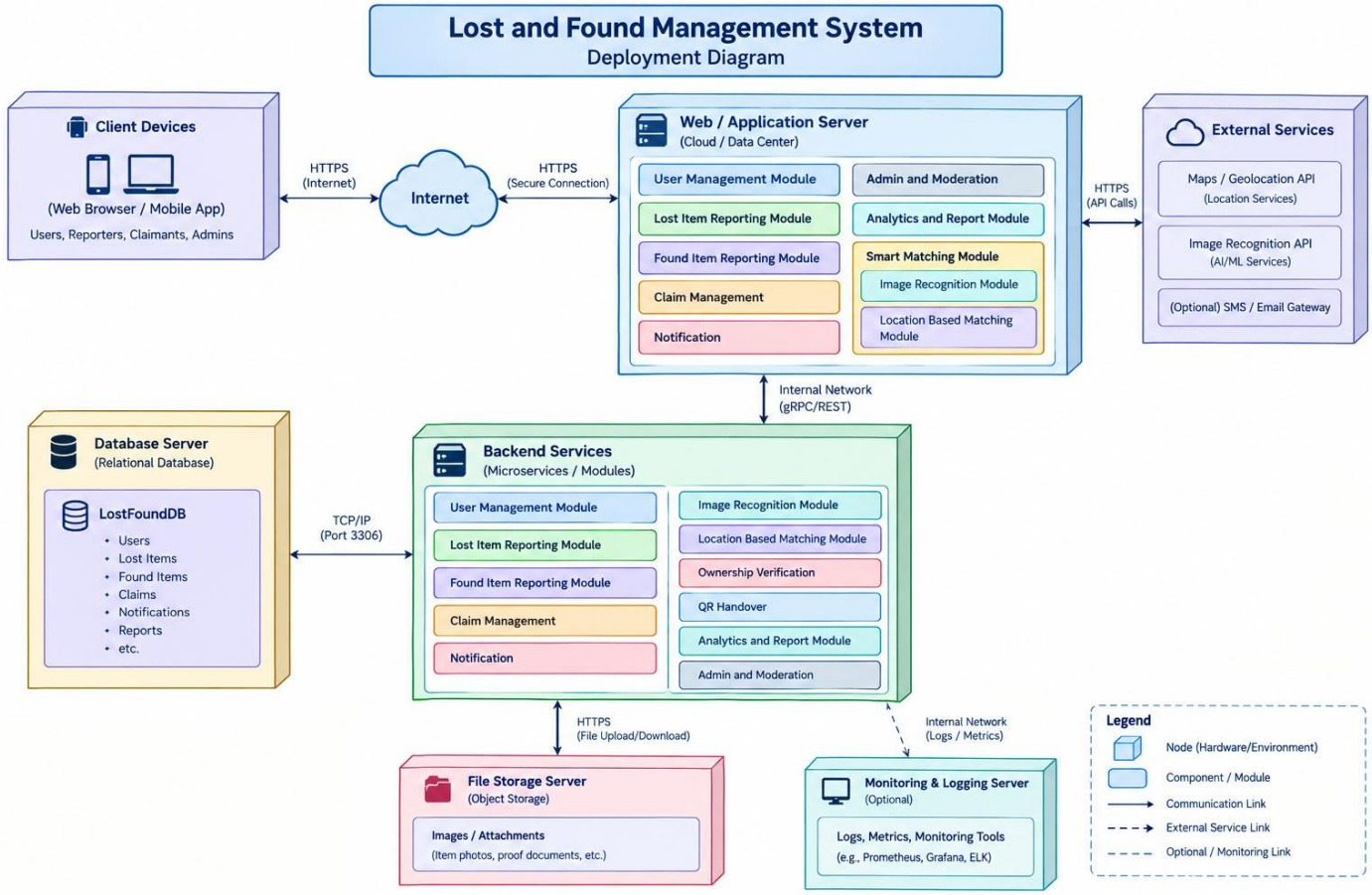


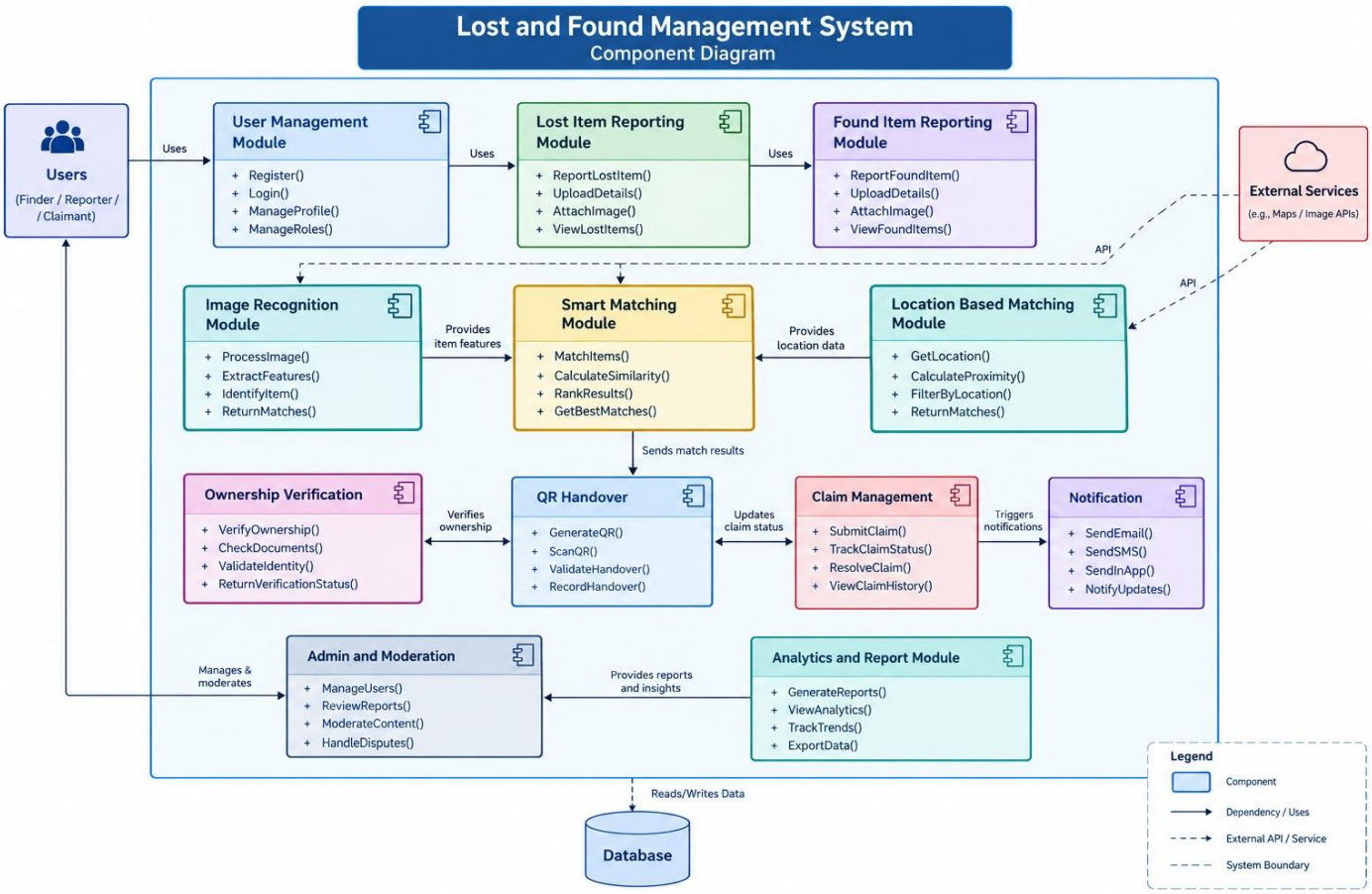
Lost and Found Management System - Simplified Sequence Diagram

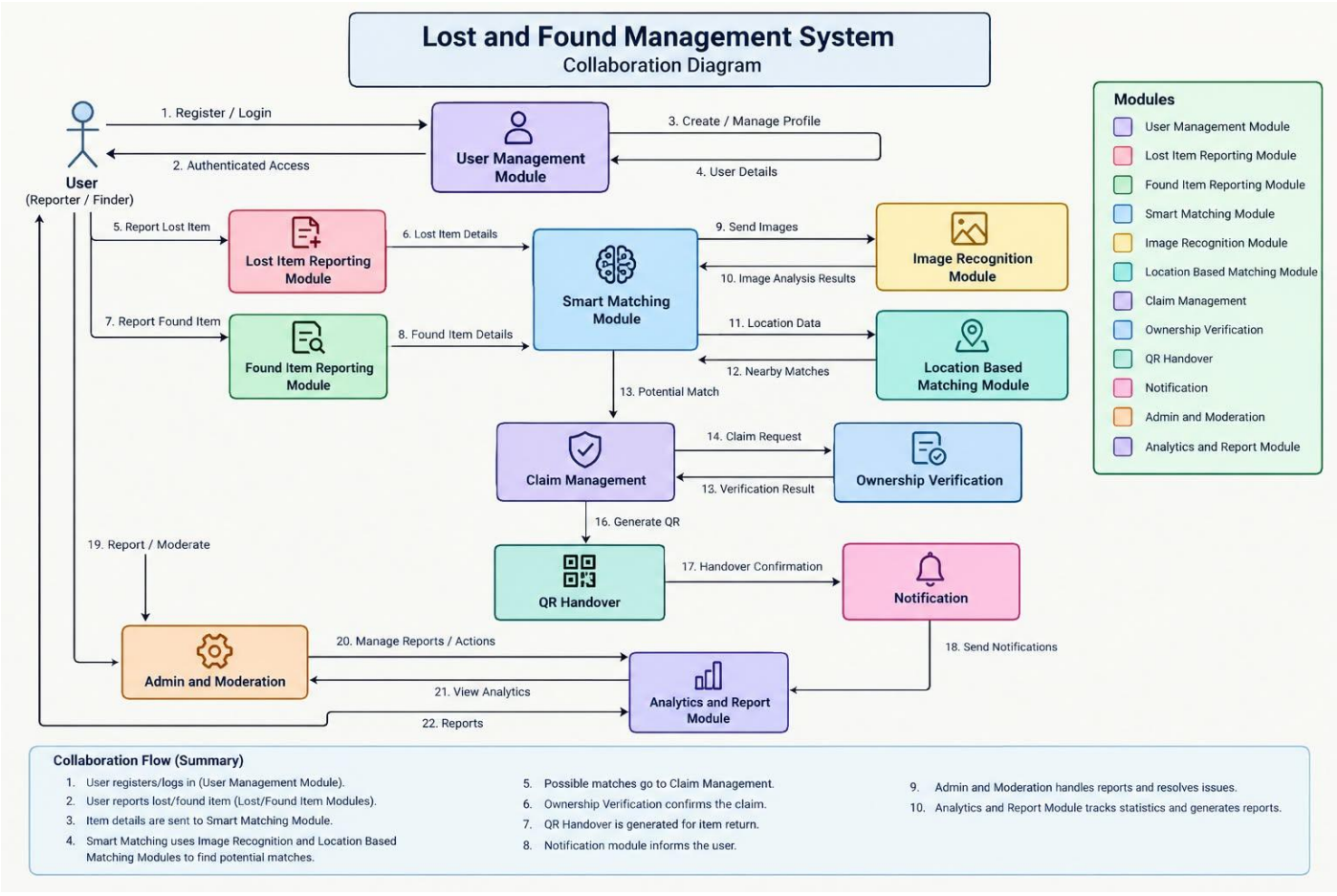


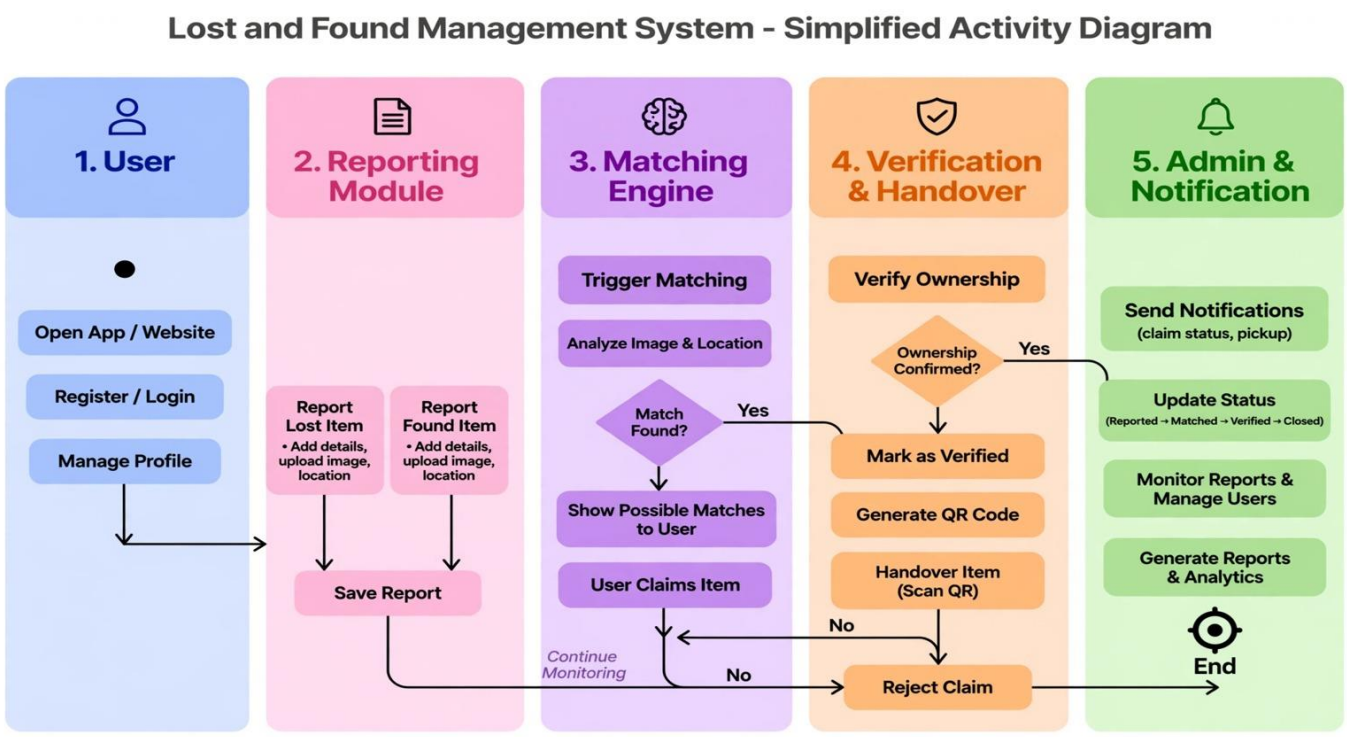
Lost and Found Management System
Package Diagram




Lost and Found Management System
Deployment Diagram
Client Devices
(Web Browser / Mobile App)
Users, Reporters, Claimants, Admins
HTTPS
(Internet)
Internet
HTTPS
(Secure Connection)
Web / Application Server
(Cloud / Data Center)
User Management Module
Admin and Moderation
Lost Item Reporting Module
Analytics and Report Module
Found Item Reporting Module
Smart Matching Module
Image Recognition Module
Location Based Matching Module
Claim Management
Notification
HTTPS
(API Calls)
External Services
Maps / Geolocation API
(Location Services)
Image Recognition API
(AI/ML Services)
(Optional) SMS / Email Gateway
Internal Network
(gRPC/REST)
Database Server
(Relational Database)
LostFoundDB
Users
Lost Items
Found Items
Claims
Notifications
Reports
etc.
TCP/IP
(Port 3306)
Backend Services
(Microservices / Modules)
User Management Module
Lost Item Reporting Module
Found Item Reporting Module
Claim Management
Notification
Image Recognition Module
Location Based Matching Module
Ownership Verification
QR Handover
Analytics and Report Module
Admin and Moderation
HTTPS
(File Upload/Download)
Internal Network
(Logs / Metrics)
File Storage Server
(Object Storage)
Images / Attachments
(Item photos, proof documents, etc.)
Monitoring & Logging Server
(Optional)
Logs, Metrics, Monitoring Tools
(e.g., Prometheus, Grafana, ELK)
Legend
Node (Hardware/Environment)
Component / Module
Communication Link
External Service Link
Optional / Monitoring Link


Lost and Found Management System
Component Diagram
Users
(Finder / Reporter / / Claimant)
Uses
User Management Module
+ Register()
+ Login()
+ ManageProfile()
+ ManageRoles()
Uses
Lost Item Reporting Module
+ ReportLostItem()
+ UploadDetails()
+ AttachImage()
+ ViewLostItems()
Uses
Found Item Reporting Module
+ ReportFoundItem()
+ UploadDetails()
+ AttachImage()
+ ViewFoundItems()
External Services
(e.g., Maps / Image APIs)
API
API
Image Recognition Module
+ ProcessImage()
+ ExtractFeatures()
+ IdentifyItem()
+ ReturnMatches()
Provides item features
Smart Matching Module
+ MatchItems()
+ CalculateSimilarity()
+ RankResults()
+ GetBestMatches()
Provides location data
Location Based Matching Module
+ GetLocation()
+ CalculateProximity()
+ FilterByLocation()
+ ReturnMatches()
Sends match results
Ownership Verification
+ VerifyOwnership()
+ CheckDocuments()
+ ValidateIdentity()
+ ReturnVerificationStatus()
Verifies ownership
QR Handover
+ GenerateQR()
+ ScanQR()
+ ValidateHandover()
+ RecordHandover()
Updates claim status
Claim Management
+ SubmitClaim()
+ TrackClaimStatus()
+ ResolveClaim()
+ ViewClaimHistory()
Triggers notifications
Notification
+ SendEmail()
+ SendSMS()
+ SendInApp()
+ NotifyUpdates()
Manages & moderates
Admin and Moderation
+ ManageUsers()
+ ReviewReports()
+ ModerateContent()
+ HandleDisputes()
Provides reports and insights
Analytics and Report Module
+ GenerateReports()
+ ViewAnalytics()
+ TrackTrends()
+ ExportData()
Reads/Writes Data
Database
Legend
Component
Dependency / Uses
External API / Service
System Boundary


Lost and Found Management System
Collaboration Diagram
User
(Reporter / Finder)
1. Register / Login
2. Authenticated Access
User Management Module
3. Create / Manage Profile
4. User Details
5. Report Lost Item
Lost Item Reporting Module
6. Lost Item Details
7. Report Found Item
Found Item Reporting Module
8. Found Item Details
Smart Matching Module
9. Send Images
10. Image Analysis Results
Image Recognition Module
11. Location Data
12. Nearby Matches
Location Based Matching Module
13. Potential Match
Claim Management
14. Claim Request
13. Verification Result
Ownership Verification
16. Generate QR
QR Handover
17. Handover Confirmation
Notification
18. Send Notifications
19. Report / Moderate
Admin and Moderation
20. Manage Reports / Actions
21. View Analytics
Analytics and Report Module
22. Reports
Modules
User Management Module
Lost Item Reporting Module
Found Item Reporting Module
Smart Matching Module
Image Recognition Module
Location Based Matching Module
Claim Management
Ownership Verification
QR Handover
Notification
Admin and Moderation
Analytics and Report Module
Collaboration Flow (Summary)
1. User registers/logs in (User Management Module).
2. User reports lost/found item (Lost/Found Item Modules).
3. Item details are sent to Smart Matching Module.
4. Smart Matching uses Image Recognition and Location Based Matching Modules to find potential matches.
5. Possible matches go to Claim Management.
6. Ownership Verification confirms the claim.
7. QR Handover is generated for item return.
8. Notification module informs the user.
9. Admin and Moderation handles reports and resolves issues.
10. Analytics and Report Module tracks statistics and generates reports.


Lost and Found Management System - Simplified Activity Diagram
1. User
Open App / Website
Register / Login
Manage Profile
2. Reporting Module
Report Lost Item
• Add details, upload image, location
Report Found Item
• Add details, upload image, location
Save Report
3. Matching Engine
Trigger Matching
Analyze Image & Location
Match Found?
Yes
Show Possible Matches to User
User Claims Item
Continue Monitoring
No
4. Verification & Handover
Verify Ownership
Ownership Confirmed?
Yes
Mark as Verified
Generate QR Code
Handover Item (Scan QR)
No
Reject Claim
5. Admin & Notification
Send Notifications
(claim status, pickup)
Update Status
(Reported → Matched → Verified → Closed)
Monitor Reports & Manage Users
Generate Reports & Analytics
End